

UNLOCKING THE ATOM

The Tiny World Inside

**AN ATOM IS LIKE
A MANGO WITH A
SEED – THE NUCLEUS!**

Science Explorer

Fission Chain Reaction



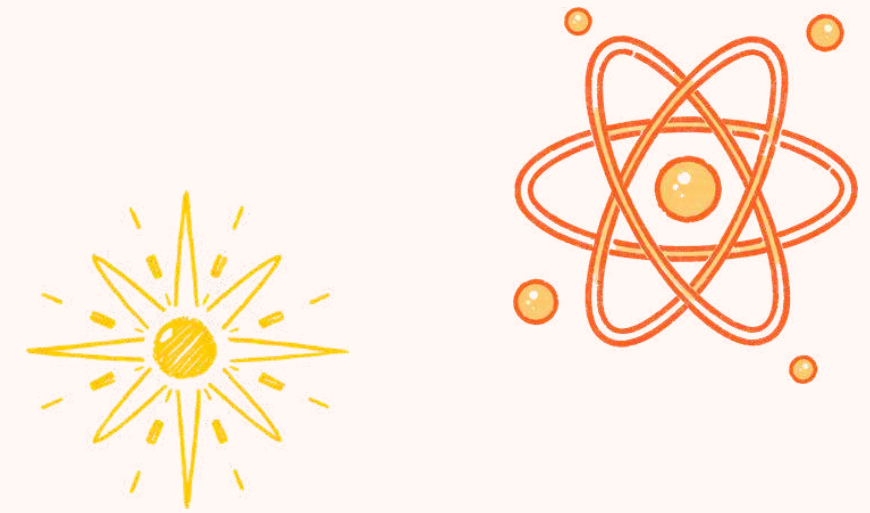
What is Nuclear Fission?

A heavy nucleus is like a matka (clay pot) that splits when hit by a tiny marble (neutron).

When the matka breaks, pieces fly out — just like a nucleus splits and releases energy and more neutrons!



The Fission Chain Reaction



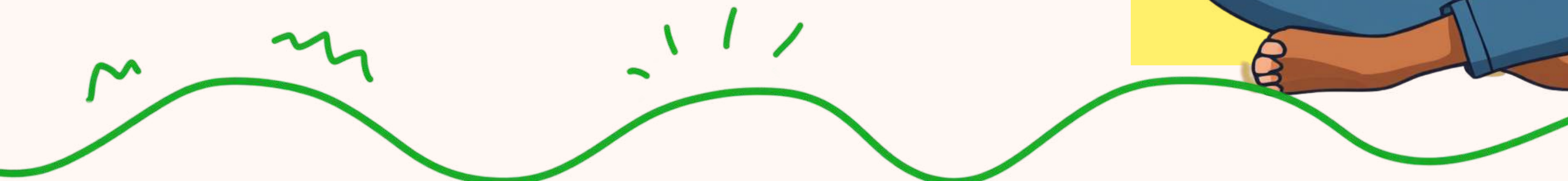
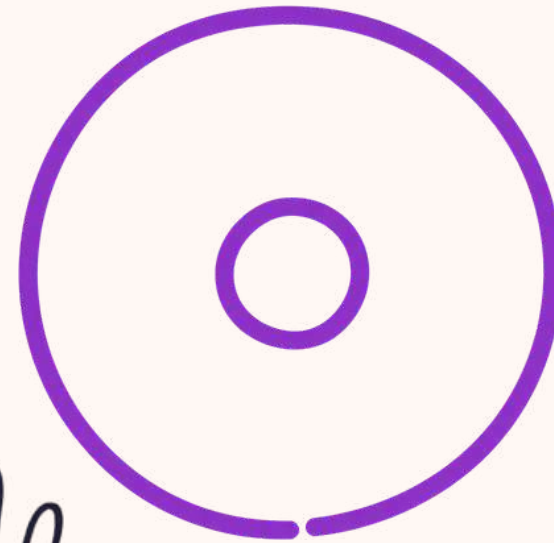
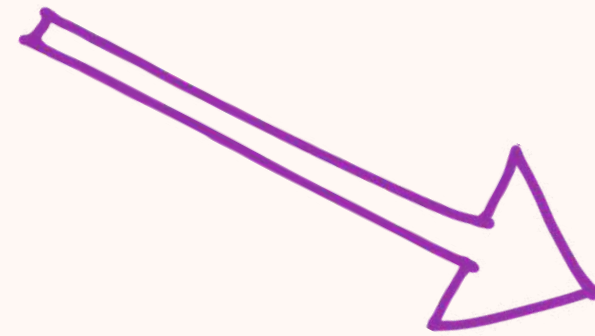
Split atoms release neutrons that hit more atoms — causing a chain reaction! Like dominoes falling one after another, each split keeps the reaction going!



Controlling the Power – Safety Brakes!

Control rods act like brakes in a nuclear reactor. When lowered into the core, they absorb extra neutrons and slow down the chain reaction — keeping everything safe and steady!

Without brakes, a car goes too fast. Without control rods, the reaction goes too fast too!





India's Nuclear Power Plants

☼ Kudankulam – Tamil Nadu

☼ Tarapur – Maharashtra

☼ Kakrapar – Gujarat

"Father of Indian Nuclear Programme"

Dr. Homi J. Bhabha 🇮🇳

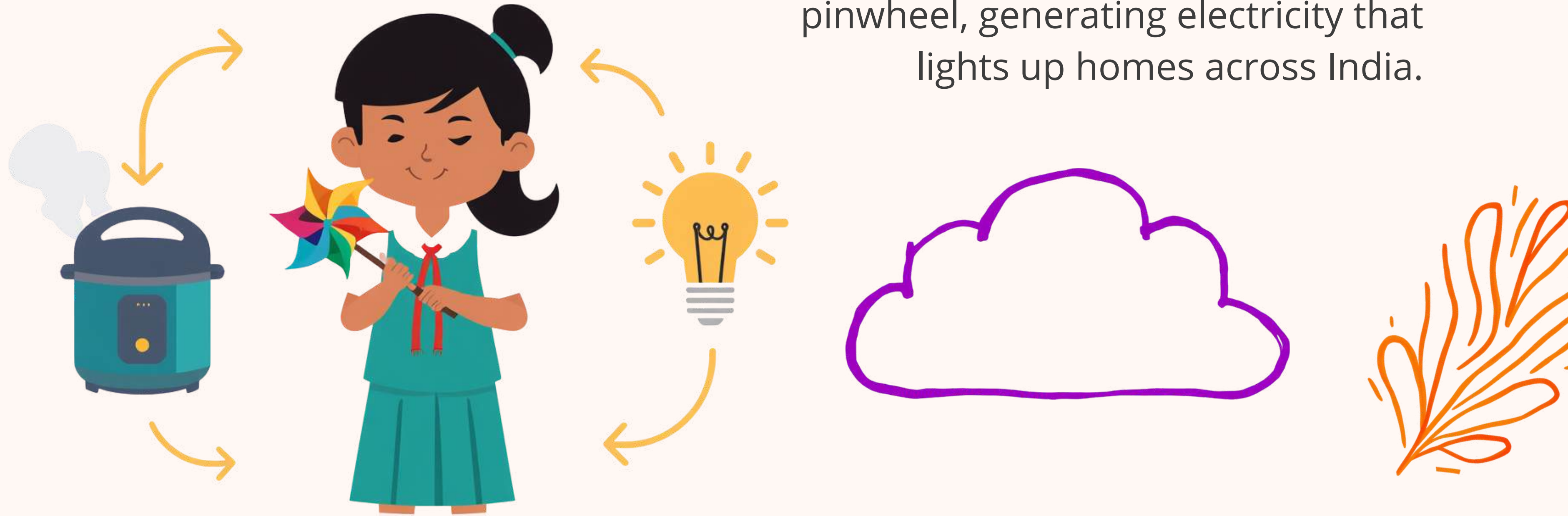
India's nuclear plants power millions of homes!



Making Electricity from Nuclear Power

Energy from fission heats water into steam — just like a pressure cooker!

The steam spins turbines like a pinwheel, generating electricity that lights up homes across India.





Today's Takeaways!

Atom = mango with a seed (nucleus)

Fission = matka breaks after marble hit

Chain reaction = dominoes falling in a row

Control rods = safety brakes for the reactor

India's nuclear plants power our homes!



Quiz Time!

1. What is the nucleus like?

a) A mango seed b) A marble c) A light bulb

2. What causes the nucleus to split?

a) A sparkler b) A marble c) A balloon

3. What do control rods do?

a) Make chain go faster b) Act as brakes c) Light bulbs



DIY Challenge!

Try making your own domino or sparkler chain reaction with friends and family!

Set up dominoes in a line and watch them fall one by one — just like a fission chain reaction!

Share your photos and stories in our next session. We can't wait to see your experiments!

